(1) a) $N(A) = n - r$
$n - 5 = 4$

$N(A^T) = m - r$
$= 7 - 5$
$= 2$

b) $m \times n = 3 \times 4$, $r = 3 \rightarrow C(A)$.

$N(A^T) = m - r$
$= 3 - 3 = 0$

2) $A \rightarrow 2 \times 3 \rightarrow r = 1 \rightarrow R(A)\ CA$

$N(A) \rightarrow 2$

$N(A^T) \rightarrow 1$

$rank \rightarrow 1$: basis $\rightarrow$ 1, 2, 4

$N(A) \rightarrow 2$: $\begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$\Rightarrow x_1 + 2x_2 + 4x_3 = 0$

$x_1 = -2x_2 - 4x_3$

$\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$

$C(A) \rightarrow 1$: basis : $[1, 2]$

$N(A^T) \rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 4 & 8 \end{bmatrix}$

$\downarrow$

1 : basis $\rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$x_1 = -2x_2$

$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

B $\rightarrow$

(3) A =

$2 \rightarrow 4$ $\quad n - r + m - r =$

20:27 Tue, 8 Apr 93%

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Given as:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

---

1 Row Space of A (Row(A))

🔹 The row space is spanned by the nonzero rows of A's row echelon form, which is U from the LU decomposition.

From U:

$$U = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

🔹 So the basis for the row space is:

$$\{\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 2 \end{bmatrix}\}$$

---

2 Column Space of A (Col(A))

🔹 We look at the pivot columns of the original matrix A, identified by the pivot positions in U.

U has pivots in:

- Column 2 (second column)
- Column 4 (fourth column)

🔹 So we take columns 2 and 4 of original matrix A:

- Column 2: $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$
- Column 4: $\begin{bmatrix} 3 & 4 & 1 \end{bmatrix}$

🔹 Therefore, the basis for the co…pace is:

Ask anything

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$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \right\}$$

---

3 Nullspace of A (Nul(A))

We solve: $A\mathbf{x} = \mathbf{0}$

From U:

$\text{Row 1: } x_2 + 2x_3 + 3x_4 + 4x_5 = 0$
$\text{Row 2: } x_4 + 2x_5 = 0$

🔸 From Row 2:
$x_4 = -2x_5$

🔸 Substitute into Row 1:

$x_2 + 2x_3 + 3(-2x_5) + 4x_5 = x_2 + 2x_3 - 6x_5 + 4x_5 = x_2 + 2x_3 - 2x_5 = 0$
$\Rightarrow x_2 = -2x_3 + 2x_5$

Let free variables:

- $x_1 = s$
- $x_3 = t$
- $x_5 = u$

Then:

- $x_2 = -2t + 2u$
- $x_4 = -2u$

So the general solution is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + u \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Ask anything

20:28 Tue, 8 Apr 93%
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🔹 Thus, basis for the nullspace:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

---

4 Left Nullspace of A (Nul(A^t))

We solve $A^T\mathbf{y} = \mathbf{0}$

Transpose of A:

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ 3 & 4 & 1 \\ 4 & 6 & 2 \end{bmatrix}$$

System:

1) $y_1 + y_2 = 0$
2) $3y_1 + 4y_2 + y_3 = 0$
3) $4y_1 + 6y_2 + 2y_3 = 0$

Step-by-step:

- From (1): $y_1 = -y_2$

Sub into (2):

$3(-y_2) + 4y_2 + y_3 = -3y_2 + 4y_2 + y_3 = y_2 + y_3 = 0 \Rightarrow y_3 = -y_2$

So:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = y_2 \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

- Therefore, basis for the left nu[illegible]e:

$\left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}$

Ask anything

4 e) $N(A)$ Dim $= 1 +$ $N(A^T)$ dim

$n - r = 1 + m - r$

$n = 1 + m$ would work.

d)

20:40 Tue, 8 Apr 93%

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Step-by-step Goal:

We want a matrix A such that:

1. $A \begin{bmatrix} 1 & 3 \end{bmatrix} = \mathbf{0}$
2. The column space of A contains $\begin{bmatrix} 3 & 1 \end{bmatrix}$

✅ Step 1: Nullspace condition

We want:

$$A \begin{vmatrix} 1 \\ 3 \end{vmatrix} = \mathbf{0}$$

That means the columns of A are linearly dependent in the following way:

$\text{col}_1 + 3 \cdot \text{col}_2 = \mathbf{0} \Rightarrow \text{col}_1 = -3 \cdot \text{col}_2$

So the two columns must satisfy that relationship. Let's write:

$$\text{col}_2 = \begin{vmatrix} x \\ y \end{vmatrix}, \quad \text{col}_1 = -3 \cdot \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} -3x \\ -3y \end{vmatrix}$$

Then the matrix becomes:

$$A = \begin{vmatrix} -3x & x \\ -3y & y \end{vmatrix}$$

✅ Step 2: Column space contains $\begin{bmatrix} 3 & 1 \end{bmatrix}$

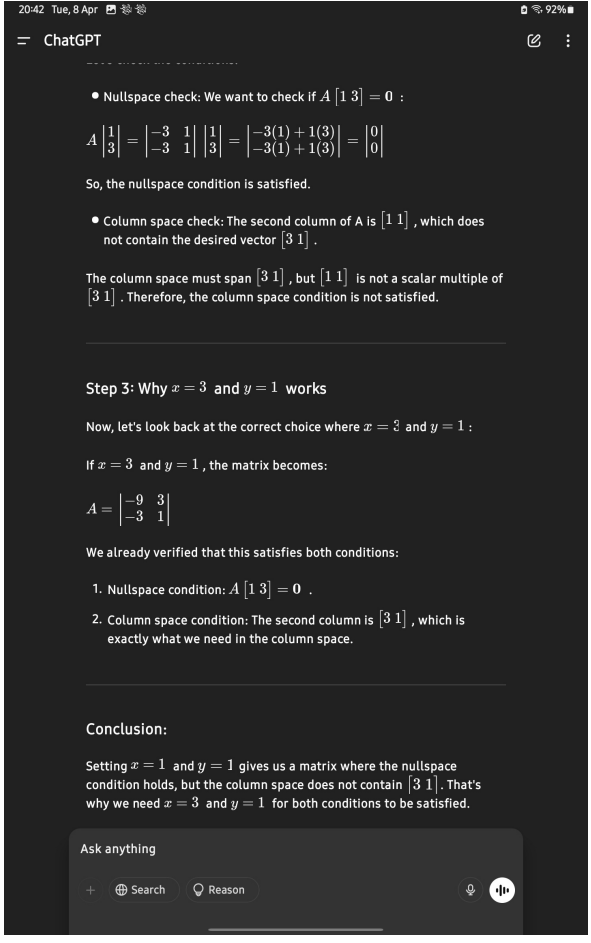
Column space = span of the columns. We want:

$$\begin{vmatrix} 3 \\ 1 \end{vmatrix} \in \text{Col}(A)$$

That means $\begin{bmatrix} 3 & 1 \end{bmatrix}$ is a scalar mult[illegible]of either column.

Let's say:

Ask anything



5

2b) $a \cdot b$

$c \quad \frac{bc}{a}$

$a \quad b$

$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$ax_1 = -bx_2$

$\therefore x_2 = \frac{a}{-b} x_1$

$\begin{bmatrix} -b \\ a \end{bmatrix}$ $x_2 = \frac{a-b}{-b}$

(22)

(23) $A = \begin{bmatrix} \quad \end{bmatrix}_{3\times 2} \begin{bmatrix} \quad \end{bmatrix}_{2\times 3}$

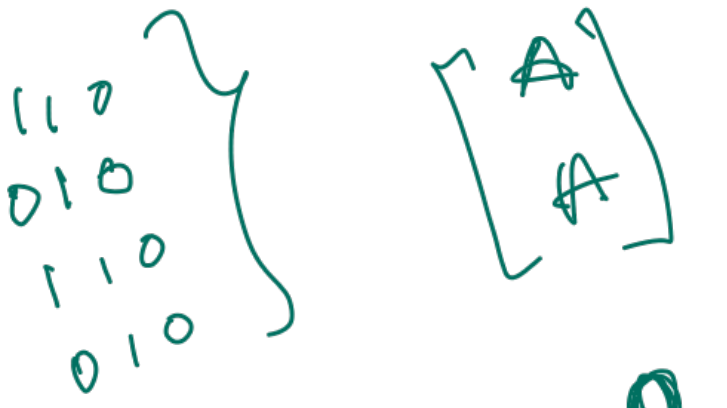
rank of 3×2 & 2×3 can't be larger the either factor.

rank ≤ 2 so, 3×3 prod not invertible

(21)

$\begin{bmatrix} 1 & 0 & \\ & & \end{bmatrix}$ $\begin{bmatrix} & & 0 \\ & & \end{bmatrix}$

not perpend.



4-2

2

2n - n